YOMNA BOUALLEGUE

Cloud Computing Engineering Student | Full-Stack Developer | DevOps
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PROFILE

3rd-year Cloud Computing engineering student at ESPRIT, seeking a 6-month final-year internship (PFE) starting January 2027 in software development, cloud or DevOps. Hands-on experience in full-stack development, Kubernetes, OpenStack and Machine Learning: designing and deploying cloud-native applications with React, Spring Boot, Node.js, Docker and Azure.

PROFESSIONAL EXPERIENCE

Internship – Full-Stack Developer & DevOps | Edora LMS July 2026 – August 2026
Edora – Analytics Portal for Moodle LMS
Technologies: *Laravel 13, PHP, PostgreSQL, Redis, Reverb, React 19, TypeScript, Docker, Nginx, Sanctum, Chart.js, Kubernetes (k3s), Jenkins*

- Developed a Laravel backend and a React 19/TypeScript SPA extending Moodle: native Moodle authentication (Web Services token, Sanctum) and scheduled synchronization of courses, grades and completions into PostgreSQL (read-only except for 3 write cases).
- Deployed a 9-service stack (Laravel, React SPA, Reverb websockets, Redis, queue worker, scheduler, PostgreSQL, Moodle, MariaDB) on a self-managed k3s cluster on Azure, with Traefik, Let's Encrypt TLS and 4 public subdomains.
- Built a Jenkins CI/CD pipeline that automatically deploys to Kubernetes on every push to main.
- Set up Prometheus/Grafana monitoring and automated backups (CronJobs) for 2 databases.
- Delivered 3 role-based dashboards (admin, teacher, student) in 2 months; Best Project Award.

Internship – Full-Stack Developer & DevOps | Teck Catalyze July 2025 – August 2025
FitLife – Fitness and Nutrition Coaching Platform
Technologies: *React Native, React.js, Node.js, Express.js, MongoDB, JWT, Docker, Nginx, Azure VM, Figma, Postman*

- Co-developed in 6 weeks a 4-role (client, coach, nutritionist, admin) fitness and nutrition coaching platform, from design to deployment.
- React Native mobile app (9 screens): workout tracking, personalized programs, BMI/BMR, geolocation, shop; React.js admin dashboard (3 screens).
- Deployed the Node.js/Express + MongoDB backend (REST API, JWT) and the React frontend behind Nginx on a 2 vCPU / 8 GB Azure VM (Docker).

ACADEMIC PROJECTS

TFAKADNI – Cloud-Native Maternal Health Platform 2025 – 2026
Technologies: *OpenStack, Kubernetes, Docker, Helm, Ansible, Prometheus, Grafana, Alertmanager, Angular, Spring Boot, MySQL, JWT*

- Deployed a 7-node OpenStack private cloud (Ubuntu Server 24.04; Keystone, Nova, Neutron/OVN, Glance, Cinder, Swift).
- Automated provisioning of a 4-node Kubernetes v1.29.15 cluster (1 control plane + 3 workers) with a single openstack stack create command (3 Heat templates + Ansible); 3-tier architecture: StatefulSets, PVCs, HPA, Ingress-nginx.
- Self-healing monitoring (Prometheus, Grafana, Alertmanager) with 3 alert rules (CrashLoop, CPU > 85%, memory > 90%): a remediator deletes the failing pod, which is recreated by the Deployment; contributed to the Angular + Spring Boot + MySQL platform (JWT/BCrypt, Scrum).

MedAssist AI – Medical Diagnosis with Machine Learning 2025 – 2026
Technologies: *Python, Scikit-learn, Pandas, NumPy, Flask, XGBoost, LightGBM, Random Forest, SVM*

- Compared 6 ML algorithms on 5 medical datasets (accuracy, precision, recall, F1-score, AUC) for early disease detection (lung cancer, brain tumor, diabetes, depression, anxiety); deployed a Flask prediction interface for physicians.

ChronoSerena – Distributed Medical Platform 2024 – 2025

- Designed a distributed platform (JavaFX desktop + Symfony web) for managing patients with chronic diseases (MVC, REST API, Singleton). Stack: JavaFX, Symfony, PHP, MySQL.

Brush & Bargain – Art Mobile App 2024 – 2025

- Developed a cross-platform mobile app for buying, selling and showcasing artworks (events, bookings, complaints). Stack: FlutterFlow, Dart, Firebase.

Kadamni – Recruitment Web Platform 2023 – 2024

- Developed a recruitment website: job posting, application management, account administration. Stack: PHP, HTML, CSS, MySQL.

AgroDesk – Farm Management + Arduino System 2023 – 2024

- Developed a C++ desktop application (OOP, Qt) for farm management, paired with an Arduino module for automated feeding scheduling. Stack: C++, Qt, Oracle DB, Arduino.

EDUCATION

Engineering Degree – Cloud Computing (3rd year) | ESPRIT, Tunis 2023 – Present

TECHNICAL SKILLS

Cloud & DevOps: OpenStack, Kubernetes, k3s, Docker, Helm, Ansible, Jenkins, Prometheus, Grafana, Alertmanager, Azure VM, AWS, Nginx, Traefik, cert-manager
Languages: Java, Python, PHP, JavaScript/TypeScript, C/C++, C#, Dart | SQL
Web & Mobile: React.js, Angular, React Native, Laravel, Spring Boot, Node.js, Express.js, Flask, Symfony, .NET, FlutterFlow
Machine Learning: Scikit-learn, Pandas, NumPy, XGBoost, LightGBM, Random Forest, SVM
Databases: PostgreSQL, MySQL, MongoDB, Redis, Oracle
Tools & Languages: Git, GitHub, Postman, Figma, UML, Agile/Scrum | French (B2), English (B2), German/Spanish (basic)